INTERNATIONAL RESEARCH WORKSHOP

Japan-Taiwan CFD Workshop

December 21-22, 2026

National Cheng Kung University | Tainan, Taiwan

Workshop Overview

The Japan-Taiwan CFD Workshop brings together researchers from Japan and Taiwan to exchange recent advances in computational fluid dynamics (CFD), high-performance computing (HPC), artificial intelligence (AI), and their applications in science and engineering.

The workshop aims to strengthen research collaboration between Japan and Taiwan, with particular emphasis on next-generation CFD software development, advanced numerical methods, high-fidelity simulations, and AI-enhanced computational modeling.

The two-day program consists of technical presentations on December 21, followed by software demonstrations and hands-on tutorials featuring CUBE and other CFD programs on the morning of December 22.

Workshop Themes

- Next-Generation HPC-CFD and GPU Computing
- Advanced Numerical Methods for Computational Fluid Dynamics
- AI-Enhanced CFD and Physics-Informed Modeling
- Turbulence, Multiphase Flows, and Fluid-Structure Interaction
- High-Fidelity CFD for Engineering and Environmental Applications
- Open CFD Software Development and Practical Applications

Organizers

Prof. ChungGang Li - Associate Professor, National Cheng Kung University

Prof. Makoto Tsubokura - Professor, Kobe University; Team Principal, RIKEN Center for Computational Science

Co-organizers

Prof. Rahul Bale - Associate Professor, Kobe University; Research Scientist, RIKEN Center for Computational Science

Prof. Ming-Jyh Chern - Professor, National Taiwan University of Science and Technology

Dr. Heng-Chuan Kan - Research Fellow, National Center for High-performance Computing

Prof. Wei-Hsiang Wang - Associate Professor, Yang Ming Chiao Tung University

DECEMBER 21 | MONDAY

Speakers & Technical Program

Technical Presentations: Recent Advances in HPC-CFD and AI

PLENARY SPEAKER

Prof. Ming-Jyh Chern

National Taiwan University of Science and Technology, Taiwan

Invited Speakers

FROM JAPAN

Makoto Tsubokura
Kobe University / RIKEN R-CCS

Rahul Bale
Kobe University / RIKEN R-CCS

Junya Onishi
RIKEN Center for Computational Science

KIM Sangwon
Affiliation to be confirmed

Alicia Maria Murga Aquino
Kobe University

FROM TAIWAN

Chih-Lun Liu
National Taiwan University

Yi-Ju Chou
National Taiwan University

Somdeb Bandopadhyay
National Cheng Kung University

Heng-Chuan Kan
National Center for High-performance Computing

December 21 Program

TIME	PROGRAM	SPEAKER / PRESENTER
09:00-09:10	Opening Remarks	Invited Guests and Organizers (TBC)
09:10-10:00	Invited Lecture	Makoto Tsubokura (Japan)
10:00-10:50	**Plenary Lecture**	**Ming-Jyh Chern (Taiwan)**
10:50-11:10	Coffee Break	
11:10-11:40	Invited Lecture	Rahul Bale (Japan)
11:40-12:10	Invited Lecture	Chih-Lun Liu (Taiwan)
12:10-12:40	Invited Lecture	Junya Onishi (Japan)
12:40-13:40	Lunch Break	
13:40-14:10	Invited Lecture	Yi-Ju Chou (Taiwan)
14:10-14:40	Invited Lecture	KIM Sangwon (Japan)
14:40-15:10	Invited Lecture	Somdeb Bandopadhyay (Taiwan)
15:10-15:30	Coffee Break	
15:30-16:00	Invited Lecture	Alicia Maria Murga Aquino (Japan)
16:00-16:30	Invited Lecture	Heng-Chuan Kan (Taiwan)
16:30-17:00	Joint Discussion and Future Collaboration	All Participants

Presentation titles will be announced upon confirmation.

DECEMBER 22 | TUESDAY

Software Demonstrations & Tutorials

Next-Generation CFD Software Demonstrations and Hands-on Tutorials

Morning program | 09:00-12:10

CUBE and next-generation CFD software: overview, updates, tutorials

December 22 Program

TIME	PROGRAM	SPEAKER / PRESENTER
09:00-09:40	Introduction to FrontFlow/blue, CUBE, and FFX: Towards Open-Source CFD Software and Community Development	RIKEN / Kobe University Team (TBC)
09:40-10:20	**Recent Developments in CUBE**	**ChungGang Li / KIM Sangwon**
10:20-10:35	Coffee Break	
10:35-11:25	CUBE Tutorial I: Geometry, Grid Generation, and Simulation Setup	TBC
11:25-12:00	CUBE Tutorial II: CFD Simulation, Execution, and Visualization	TBC
12:00-12:10	Closing Remarks	Organizers

END OF WORKSHOP **12:10**

Open-Source Software and Community Development

The introductory session will present the capabilities of FrontFlow/blue, CUBE, and FFX, highlight their planned open-source development and the release of reusable software components, and encourage researchers and students to adopt and contribute to the emerging HPC-CFD software ecosystem.

PROGRAM NOTES

Tutorial topics, software availability, and technical requirements are subject to confirmation.